2 A non-uniform rigid body is perfectly balanced on a smooth, fixed support.

Which of the following statements correctly describes the physical state of the body in this position?

A The body is in equilibrium because the total mass on the left side of the support is exactly equal to the total mass on the right side of the support.

B The support must be located at the geometric centre of the rigid body to ensure the resultant torque is zero.

C The line of action of the body's weight passes directly through the support, ensuring the resultant torque is zero.

D The body is in equilibrium because the upward force exerted by the body on the support is equal to the weight of the body.